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RESEARCH ARTICLE

On the team-maxmin equilibria

Takuya Iimura[1]

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Abstract
In this note we extend the result of von Stengel and Koller (Games Econ Behav 21:309–321, 1997) to infinite games. Specifically, we show that every infinite zero-sum game between a team and an adversary admits a pure-strategy team-maxmin equilibrium when (i) the strategy sets are compact, convex subsets of (possibly infinite-dimensional) topological vector spaces, and (ii) the payoff function is bounded, upper-semicontinuous on the team's strategy-profile set and concave in each team member's strategy, while being lower-semicontinuous and convex in the adversary's strategy. Because the vector spaces may have arbitrary dimension, we obtain the following corollary: a mixed-strategy team-maxmin equilibrium exists whenever (i) the strategy sets are compact subsets of metric spaces, and (ii) the payoff function is bounded, measurable, upper-semicontinuous on the team's strategy-profile set, and lower-semicontinuous on the adversary's strategy set. The proof employs Sion's minimax theorem.

Keywords Team-maxmin equilibrium · Sion's minimax theorem · Zero-sum games · Existence of equilibrium

JEL Classification C72 (Noncooperative game)

1 Introduction

Von Stengel and Koller (1997) show that every *finite* zero-sum game between a *team* and an *adversary* possesses a *team-maxmin equilibrium*—an equilibrium in which the team adopts its maxmin strategy profile. Here, the *team* comprises several players who share identical payoffs and make a collective strategy choice. The team's maxmin strategy profile is a collection of (possibly mixed) strategies, one for each team member, that guarantees the highest worst-case payoff for the team. Crucially, these strategies are *not* correlated; hence the equilibrium is an equilibrium of the original multi-player

✉ Takuya Iimura
t.iimura@tmu.ac.jp

1 School of Business Administration, Tokyo Metropolitan University, Tokyo 192-0397, Japan

zero-sum game. Moreover, among all equilibria, it yields the greatest payoff to the team.

We extend this result to *infinite* games. Specifically, we prove that an infinite zero-sum game between a team and an adversary admits a *pure-strategy* team-maxmin equilibrium whenever (i) the strategy sets are compact, convex subsets of (possibly infinite-dimensional) topological vector spaces; and (ii) the payoff function is bounded, upper-semicontinuous on the team's strategy-profile set and concave in each team member's strategy, while lower-semicontinuous and convex in the adversary's strategy. Because the underlying vector spaces may have arbitrary dimension, we obtain the following corollary: a mixed-strategy team-maxmin equilibrium exists whenever (i) the strategy sets are compact subsets of metric spaces; and (ii) the payoff function is bounded, measurable, upper-semicontinuous on the team's strategy-profile set, and lower-semicontinuous on the adversary's strategy set.

For both finite and infinite games, the existence of an equilibrium in mixed strategies can be established using a fixed-point argument (Nash 1950; Glicksberg 1952). A notable aspect of von Stengel and Koller's result is that they prove the existence of an equilibrium of a special kind without invoking any fixed-point theorem; instead, they use linear programming (LP) duality. In this paper, we apply Sion's minimax theorem (Sion 1958). We emphasize that, although Sion's original proof relies on the KKM lemma and Helly's theorem, an elementary proof is available (Komiya 1988). Our framework subsumes that of von Stengel and Koller and extends the theory of team-maxmin equilibria from the mixed-strategy equilibria in finite games to the pure- and mixed-strategy equilibria in infinite games.

The remainder of the paper is organized as follows. Section 2 presents the formal setup and notation. Section 3 contains the main existence results. Section 4 concludes with some remarks.

2 Preliminaries

Denote by $G = (X_1, \dots, X_n, Y, u)$ an $(n+1)$-player zero-sum game between n members of a team and an adversary to the team, where $X_1, \dots, X_n, Y$ are non-empty strategy sets of players, X_i for the team member i, $i = 1, \dots, n$, Y for the adversary, and $u\colon X \times Y \to \mathbb{R}$ is the payoff function of team member, where $X = X_1 \times \cdots \times X_n$, the set of all strategy profiles of the team. Every member of the team has the identical payoff function u, and the adversary's payoff is given by the negative of n times of u, i.e., by $-nu\colon X \times Y \to \mathbb{R}$.

The following definitions are given by von Stengel and Koller (1997). First, a *team-maxmin strategy profile* is an element $\underline{x} \in X$ such that

$$\min_{y \in Y} u(\underline{x}, y) = \max_{x \in X} \min_{y \in Y} u(x, y), \tag{1}$$

namely, a strategy profile of the team that ensures the highest worst-case payoff for the team. Second, as a notion of equilibria of this $(n+1)$-player zero-sum game, a *team-maxmin equilibrium* is a Nash equilibrium $(\underline{x}, \overline{y}) \in X \times Y$ of G, where $\underline{x} \in X$

is a team-maxmin strategy profile and $\overline{y}$ is a strategy of the adversary; namely, a pair of a team-maxmin strategy profile $\underline{x}$ and some strategy $\overline{y}$ of the adversary satisfying

$$u(\underline{x}_i, \underline{x}_{-i}, \overline{y}) \geq u(x_i, \underline{x}_{-i}, \overline{y}) \; \forall x_i \in X_i, \; \forall i, \text{ and } u(\underline{x}, \overline{y}) \leq u(\underline{x}, y) \; \forall y \in Y, \quad (2)$$

where $\underline{x}_{-i} = (\underline{x}_1, \ldots, \underline{x}_{i-1}, \underline{x}_{i+1}, \ldots, \underline{x}_n) \in \prod_{j \neq i} X_j$, as usual.

In von Stengel and Koller (1997), the game G was a mixed extension of a finite game, in which a team-maxmin strategy profile is guaranteed to exist. They showed, in this setting:

Theorem 2.1 (von Stengel and Koller 1997) *Any team-maxmin strategy profile is part of a team-maxmin equilibrium.*

We will extend the domain of this result in the next section. To this end, we will use the following theorem.

Theorem 2.2 (Sion 1958) *Let S and T be compact and convex subsets of two topological vector spaces and $f : S \times T \to \mathbb{R}$. If f is upper-semicontinuous (usc) and quasiconcave on S, and if f is lower-semicontinuous (lsc) and quasiconvex on T, then*

$$\max_{s \in S} \min_{t \in T} f(s, t) = \min_{t \in T} \max_{s \in S} f(s, t). \quad (3)$$

Here, f is quasiconcave on S if the set $\{s \in S \mid f(s, t) \geq \alpha\}$ is convex for every $\alpha \in \mathbb{R}$ given any $t \in T$; quasiconvex on T if the set $\{t \in T \mid f(s, t) \leq \alpha\}$ is convex for every $\alpha \in \mathbb{R}$ given any $s \in S$. Under Eq. (3) (hereafter called the *maxmin equality*), we have

$$f(s, \overline{t}) \leq f(\underline{s}, \overline{t}) \leq f(\underline{s}, t) \quad \forall s \in S, \forall t \in T, \quad (4)$$

with a maxmin strategy $\underline{s} \in S$ such that $\min_t f(\underline{s}, t) = \max_s \min_t f(s, t)$ and a minmax strategy $\overline{t} \in T$ such that $\max_s f(s, \overline{t}) = \min_t \max_s f(s, t)$. Letting $\Gamma = (S, T, f)$ be a two-player zero-sum game, Eq. (4) says that $(\underline{s}, \overline{t}) \in S \times T$ is an *equilibrium* of Γ. Conversely, if Γ has an equilibrium $(s^*, t^*) \in S \times T$, then s^* is a maxmin strategy and t^* is a minmax strategy (Osborne and Rubinstein 1994, Proposition 22.2). Thus, for a two-player zero-sum game *possessing* an equilibrium, a pair of maxmin and minmax strategies is equivalent to an equilibrium.

Recall that if f is *concave* on S, i.e., if $f((1-\theta)s + \theta s', t) \geq (1-\theta) f(s, t) + \theta f(s', t)$ for any $s, s' \in S$ and $\theta \in [0, 1]$ given any $t \in T$, then it is quasiconcave on S; if f is *convex* on T, i.e., if $f(s, (1-\theta)t + \theta t') \leq (1-\theta) f(s, t) + \theta f(s, t')$ for any $t, t' \in T$ and $\theta \in [0, 1]$ given any $s \in S$, then it is quasiconvex on T.

3 Main results

Let $G = (X_1, \ldots, X_n, Y, u)$ be a zero-sum game between a team and an adversary, and let $X = X_1 \times \cdots \times X_n$. We consider the following set of conditions.

Assumption 3.1 1. $X_1, \ldots, X_n, Y$ are compact, convex subsets of (possibly infinite-dimensional) topological vector spaces.
2. u is bounded, usc on X and concave on each X_i, $i = 1, \ldots, n$, while lsc and convex on Y.

Note that X, endowed with the product topology, is also compact, and convex.

Theorem 3.1 *Under Assumption 3.1, a team-maxmin strategy profile exists, and any team-maxmin strategy profile is part of a team-maxmin equilibrium.*

The proof of this theorem is almost identical to the proof given in von Stengel and Koller (1997), except that we use Sion's theorem, instead of duality of LP.

Proof For any $x \in X$, $u(x, \cdot)$ is lsc on compact Y. Hence, $\min_{y \in Y} u(x, y)$ exists for any $x \in X$. Consider the function $\min_{y \in Y} u(\cdot, y)$, the pointwise minimum of the functions $u(\cdot, y)$, $y \in Y$. It is usc, because $u(\cdot, y)$ is usc for any $y \in Y$ and the pointwise minimum of usc functions is usc. Hence, it has a maximizer in compact X, which is a team-maxmin strategy profile. Let $\underline{x} \in X$ be an arbitrary team-maxmin strategy profile, and define $v_i : X_i \times Y \to \mathbb{R}$ by $v_i(x_i, y) = u(x_i, \underline{x}_{-i}, y)$ for each $i = 1, \ldots, n$ (v_i depends on $\underline{x}$, but we omit it for notational simplicity). Then their sum $\sum_i v_i = \sum_{i=1}^n v_i$ is usc and concave on X, and lsc and convex on Y. By Sion's theorem, it holds that

$$\max_{x \in X} \min_{y \in Y} \sum_i v_i(x_i, y) = \min_{y \in Y} \max_{x \in X} \sum_i v_i(x_i, y). \tag{5}$$

Consider a two-player zero-sum game $\Gamma_1 = (X, Y, \sum_i v_i)$, in which the maxmin equality Eq. (5) holds true. We claim that

$$\max_{x \in X} \min_{y \in Y} \sum_i v_i(x_i, y) = \min_{y \in Y} \sum_i v_i(\underline{x}_i, y), \tag{6}$$

i.e., the team-maxmin strategy profile $\underline{x}$ is a maxmin strategy of x-player in Γ_1. To see this, notice that $\max_x \min_y \sum_i v_i(x_i, y) \geq \min_y \sum_i v_i(\underline{x}_i, y)$, and suppose to the contrary that

$$\max_{x \in X} \min_{y \in Y} \sum_i v_i(x_i, y) > \min_{y \in Y} \sum_i v_i(\underline{x}_i, y). \tag{7}$$

If $\hat{x} \in X$ is a maximizer of the left-hand side, then $\min_y \sum_i v_i(\hat{x}_i, y) > \min_y \sum_i v_i(\underline{x}_i, y)$. If $\hat{y} \in Y$ and $y' \in Y$ are minimizers of $\sum_i v_i(\hat{x}_i, \cdot) = \sum_i u(\hat{x}_i, \underline{x}_{-i}, \cdot)$ and $\sum_i v_i(\underline{x}_i, \cdot) = nu(\underline{x}, \cdot)$, respectively, then

$$\sum_i u(\hat{x}_i, \underline{x}_{-i}, \hat{y}) > nu(\underline{x}, y'). \tag{8}$$

Of course y' also minimizes $v_1(\underline{x}_1, \cdot) = \cdots = v_n(\underline{x}_n, \cdot) = u(\underline{x}, \cdot)$. Now, since u is concave on each X_i, $i = 1, \ldots, n$, we have that, for any $\epsilon \in]0, 1[$ and $y \in Y$,

$$\begin{aligned} u((1-\epsilon)\underline{x}+\epsilon\hat{x}, y) &= u((1-\epsilon)\underline{x}_1+\epsilon\hat{x}_1, \ldots, (1-\epsilon)\underline{x}_n+\epsilon\hat{x}_n, y) \\ &\geq (1-\epsilon)u(\underline{x}_1, (1-\epsilon)\underline{x}_2+\epsilon\hat{x}_2, \ldots, (1-\epsilon)\underline{x}_n+\epsilon\hat{x}_n, y) \\ &\quad +\epsilon u(\hat{x}_1, (1-\epsilon)\underline{x}_2+\epsilon\hat{x}_2, \ldots, (1-\epsilon)\underline{x}_n+\epsilon\hat{x}_n, y) \\ &\geq \cdots \\ &\geq (1-\epsilon)^n u(\underline{x}, y)+\epsilon(1-\epsilon)^{n-1}\sum_i u(\hat{x}_i, \underline{x}_{-i}, y)+\epsilon^2 A(\epsilon, \hat{x}, \underline{x}, y) \\ &= u(\underline{x}, y)-n\epsilon u(\underline{x}, y)+\epsilon\sum_i u(\hat{x}_i, \underline{x}_{-i}, y)+\epsilon^2 B(\epsilon, \hat{x}, \underline{x}, y), \end{aligned}$$

where $A(\epsilon, \hat{x}, \underline{x}, y)$ and $B(\epsilon, \hat{x}, \underline{x}, y)$ are expressions for $\epsilon \in]0, 1[$ and $y \in Y$ that are bounded due to the boundedness of u. [1]Moreover, for $\epsilon \in]0, \frac{1}{n}[$, we have $(1-n\epsilon)u(\underline{x}, y) \geq (1-n\epsilon)u(\underline{x}, y')$ and $\sum_i u(\hat{x}_i, \underline{x}_{-i}, y) \geq \sum_i u(\hat{x}_i, \underline{x}_{-i}, \hat{y})$ for any

[1] Precise forms of $A(\epsilon, \hat{x}, \underline{x}, y)$ and $B(\epsilon, \hat{x}, \underline{x}, y)$ are as follows. Let $N = \{1, \ldots, n\}$ and $x_I := (x_i \mid i \in I)$ for $I \subseteq N$. Now, since

$$u((1-\epsilon)\underline{x}+\epsilon\hat{x}, y) \geq \sum_{k=0}^{n} \epsilon^k (1-\epsilon)^{n-k} \sum_{I\subseteq N, |I|=k} u(\hat{x}_I, \underline{x}_{-I}, y)$$

$$= (1-\epsilon)^n u(\underline{x}, y)+\epsilon(1-\epsilon)^{n-1}\sum_i u(\hat{x}_i, \underline{x}_{-i}, y)+\sum_{k=2}^{n}\epsilon^k(1-\epsilon)^{n-k}\sum_{I\subseteq N, |I|=k} u(\hat{x}_I, \underline{x}_{-I}, y),$$

we set

$$A(\epsilon, \hat{x}, \underline{x}, y) = \sum_{k=2}^{n}\epsilon^{k-2}(1-\epsilon)^{n-k}\sum_{I\subseteq N, |I|=k} u(\hat{x}_I, \underline{x}_{-I}, y).$$

Since

$$(1-\epsilon)^n = \sum_{k=0}^{n}\binom{n}{k}(-\epsilon)^k = 1-n\epsilon+\sum_{k=2}^{n}\binom{n}{k}(-\epsilon)^k$$

and

$$\begin{aligned} \epsilon(1-\epsilon)^{n-1} &= \epsilon\left(1-(n-1)\epsilon+\sum_{k=2}^{n-1}\binom{n-1}{k}(-\epsilon)^k\right) \\ &= \epsilon+\epsilon^2\left(1-n+\epsilon\sum_{k=2}^{n-1}\binom{n-1}{k}(-\epsilon)^{k-2}\right), \end{aligned}$$

we have

$$\begin{aligned} (1-\epsilon)^n u(\underline{x}, y)+\epsilon(1-\epsilon)^{n-1}\sum_i u(\hat{x}_i, \underline{x}_{-i}, y) &= u(\underline{x}, y)-n\epsilon u(\underline{x}, y)+\epsilon\sum_i u(\hat{x}_i, \underline{x}_{-i}, y) \\ &+\epsilon^2\sum_{k=2}^{n}\binom{n}{k}(-\epsilon)^{k-2}u(\underline{x}, y)+\epsilon^2\left(1-n+\epsilon\sum_{k=2}^{n-1}\binom{n-1}{k}(-\epsilon)^{k-2}\right)\sum_i u(\hat{x}_i, \underline{x}_{-i}, y), \end{aligned}$$

so we set

$$B(\epsilon, \hat{x}, \underline{x}, y) = \sum_{k=2}^{n}\binom{n}{k}(-\epsilon)^{k-2}u(\underline{x}, y)+\left(1-n+\epsilon\sum_{k=2}^{n-1}\binom{n-1}{k}(-\epsilon)^{k-2}\right)\sum_i u(\hat{x}_i, \underline{x}_{-i}, y)+A(\epsilon, \hat{x}, \underline{x}, y).$$

$y \in Y$ by the definition of y' and $\hat{y}$, so

$$u((1-\epsilon)\underline{x}+\epsilon\hat{x}, y) \geq u(\underline{x}, y') - n\epsilon u(\underline{x}, y') + \epsilon \sum_i u(\hat{x}_i, \underline{x}_{-i}, \hat{y}) \\ +\epsilon^2 B(\epsilon, \hat{x}, \underline{x}, y) \tag{9}$$

for any $y \in Y$ given any $\epsilon \in]0, \frac{1}{n}[$. Now, for any $\epsilon \in]0, \frac{1}{n}[$, let $y^\epsilon \in Y$ be such that

$$u((1-\epsilon)\underline{x}+\epsilon\hat{x}, y^\epsilon) = \min_{y\in Y} u((1-\epsilon)\underline{x}+\epsilon\hat{x}, y). \tag{10}$$

Then, by Eq. (9),

$$u((1-\epsilon)\underline{x}+\epsilon\hat{x}, y^\epsilon) \geq u(\underline{x}, y') + \epsilon \left(\sum_i u(\hat{x}_i, \underline{x}_{-i}, \hat{y}) - nu(\underline{x}, y')\right) \\ +\epsilon^2 B(\epsilon, \hat{x}, \underline{x}, y^\epsilon). \tag{11}$$

Since the expression in the parenthesis is positive by Eq. (8), and since $B(\epsilon, \hat{x}, \underline{x}, y^\epsilon)$ is bounded, we have for sufficiently small ϵ,

$$\min_{y\in Y} u((1-\epsilon)\underline{x}+\epsilon\hat{x}, y) = u((1-\epsilon)\underline{x}+\epsilon\hat{x}, y^\epsilon) > u(\underline{x}, y') = \min_{y\in Y} u(\underline{x}, y), \tag{12}$$

i.e., $x' := (1-\epsilon)\underline{x} + \epsilon\hat{x}$ ensures a higher worst-case payoff for the team than $\underline{x}$ does, contradicting that $\underline{x}$ is a team-maxmin strategy profile. Hence Eq. (6). Having established that $\underline{x}$ is a maxmin strategy of x-player in Γ_1 satisfying the maxmin equality Eq. (5), let $\overline{y} \in Y$ be a minmax strategy of y-player in Γ_1. Then the pair $(\underline{x}, \overline{y})$ constitutes an equilibrium of Γ_1. That $\sum_i v_i(\underline{x}_i, \overline{y}) \geq \sum_i v_i(x_i, \overline{y})$ for any $x \in X$ means that $u(\underline{x}_i, \underline{x}_{-i}, \overline{y}) \geq u(x_i, \underline{x}_{-i}, \overline{y})$ for any $x_i \in X_i$, for every $i = 1, \ldots, n$. That $\sum_i v_i(\underline{x}_i, \overline{y}) \leq \sum_i v_i(\underline{x}, y)$ for any $y \in Y$ means that $u(\underline{x}, \overline{y}) \leq u(\underline{x}, y)$ for any $y \in Y$. Hence, $(\underline{x}, \overline{y})$ is a team-maxmin equilibrium of G. □

The *mixed extension* of a game $G = (X_1, \ldots, X_n, Y, u)$ is a game $\overline{G} = (M_1, \ldots, M_n, N, \tilde{u})$, where $M_1, \ldots, M_n, N$ are the sets of *mixed* strategies of players (the sets of all probability measures on $X_1, \ldots, X_n, Y$, respectively), M_i for the team member i, $i = 1, \ldots, n$, and N for the adversary, and $\tilde{u}\colon M_1 \times \cdots \times M_n \times N \to \mathbb{R}$ is the expected payoff of a team member given by

$$\tilde{u}(\mu_1, \ldots, \mu_n, \nu) = \int_{X_1\times\cdots\times X_n\times Y} u(x_1, \ldots, x_n, y) d(\mu_1 \times \cdots \times \mu_n \times \nu), \tag{13}$$

where $\mu_1 \times \cdots \times \mu_n \times \nu$ is the product measure induced from $(\mu_1, \ldots, \mu_n, \nu)$ via $(\mu_1 \times \cdots \times \mu_n \times \nu)(S_1 \times \cdots \times S_n \times T) = \mu_1(S_1)\ldots\mu_n(S_n)\nu(T)$ (with $S_1 \subseteq X_1, \ldots, S_n \subseteq X_n, T \subseteq Y$). To be precise, if $\Delta(A)$ is the set of all probability measures on A,[2] then $M_i = \Delta(X_i)$, $i = 1, \ldots, n$, $N = \Delta(Y)$, $(\mu_1, \ldots, \mu_n, \nu) \in \Delta(X_1) \times \cdots \times \Delta(X_n) \times \Delta(Y)$, and $\mu_1 \times \cdots \times \mu_n \times \nu \in \Delta(X_1 \times \cdots \times X_n \times Y)$. Now, when we consider $\overline{G}$, we do not need any convexity assumptions for the sets of *pure* strategies $X_1, \ldots, X_n, Y$ and the payoff u of G. The sets $X_1, \ldots, X_n, Y$ even need not be subsets of vector spaces, but, in order to use probability measures on them, we assume that they are subsets of metric spaces. Thus, we assume the following conditions for G. Let $X = X_1 \times \cdots \times X_n$:

Assumption 3.2 1. $X_1, \ldots, X_n, Y$ are compact subsets of metric spaces.
2. u is bounded, measurable, usc on X, and lsc on Y.

By the boundedness and measurability, u is $(\mu_1 \times \cdots \times \mu_n \times \nu)$-integrable for any $(\mu_1, \ldots, \mu_n, \nu) \in M_1 \times \cdots \times M_n \times N$. The boundedness of u also ensures the boundedness of $\tilde{u}$ (we have $-\infty < \inf u \leq \tilde{u} \leq \sup u < +\infty$). Inherently, the sets $M_1, \ldots, M_n, N$ are convex subsets of vector spaces. If $X_1, \ldots, X_n, Y$ are compact subsets of metric spaces, then $M_1, \ldots, M_n, N$ are compact under the weak* topologies (Aliprantis and Border 2006, Theorem 15.11). Let $\mu = \mu_1 \times \cdots \times \mu_n \in \Delta(X)$. By Fubini's theorem, the $(\mu \times \nu)$-integrability implies

$$\begin{aligned} \tilde{u}(\mu_1, \ldots, \mu_n, \nu) &= \int_{X_1} \cdots \int_{X_n} \int_Y u(x_1, \ldots, x_n, y) d\nu d\mu_n \ldots d\mu_1 \\ &= \int_X \left[\int_Y u(x, y) d\nu \right] d\mu = \int_Y \left[\int_X u(x, y) d\mu \right] d\nu, \end{aligned} \tag{14}$$

where $x = (x_1, \ldots, x_n)$. Thus, $\tilde{u}$ is affine on each of $M_1, \ldots, M_n, N$. Let $F(x) = \int_Y u(x, y) d\nu$, $G(y) = \int_X u(x, y) d\mu$, and $M = M_1 \times \cdots \times M_n$, endowing the product topology. The map $(\mu_1, \ldots, \mu_n) \mapsto \mu$ from M to $\Delta(X)$ with weak* topology is continuous (Aliprantis et al. 2006, Lemma 3.4). The map $\mu \mapsto \int_X F(x) d\mu$ from $\Delta(X)$ to $\mathbb{R}$ is usc if the bounded function F is usc, and the map $\nu \mapsto \int_Y G(y) d\nu$ from $N = \Delta(Y)$ to $\mathbb{R}$ is lsc if the bounded function G is lsc (Aliprantis and Border 2006, Theorem 15.5). One can prove that F is usc and G is lsc.[3] Hence, $\tilde{u}$ is usc on M (taking the composition) and lsc on N.

We thus have:

1. $M_1, \ldots, M_n, N$ are compact, convex subsets of topological vector spaces.
2. $\tilde{u}$ is bounded, usc on M, affine on each M_i, $i = 1, \ldots, n$, and lsc and affine on N.

Since affine function is both concave and convex, we have the following corollary.

Corollary 3.1 *Under Assumption 3.2, a mixed-strategy team-maxmin strategy profile exists, and any team-maxmin strategy profile is part of a team-maxmin equilibrium.*

[2] In our setting, A is a (metrizable) topological space, and we mean by "probability measures on A" the probability measures defined on the Borel σ-algebra of A.

[3] Since X and Y are metrizable, we can work with sequences. Since u is bounded, we have, if $x^n \to x$, then $\overline{\lim} F(x^n) = \overline{\lim} \int_Y u(x^n, y) d\nu \leq \int_Y \overline{\lim}\, u(x^n, y) d\nu \leq \int_Y u(x, y) d\nu = F(x)$ by "reverse" Fatou's lemma and usc of u in x. Hence, F is usc. If $y^n \to y$, then $\underline{\lim}\, G(y^n) = \underline{\lim} \int_X u(x, y^n) d\mu \geq \int_X \underline{\lim}\, u(x, y^n) d\mu \geq \int_X u(x, y) d\mu = G(y)$ by Fatou's lemma and lsc of u in y. Hence, G is lsc.

The mixed-strategy team-maxmin equilibria of finite games can also be viewed as an example of this corollary: If G is a finite game, then the finite strategy sets $X_1, \dots, X_n, Y$ are subsets of discrete metric spaces. They are compact in discrete topology, and the bounded payoff u on a finite set $X \times Y$ is usc on X and lsc on Y under the same topology; also u is trivially measurable. Thus, it satisfies Assumption 3.2.

4 Concluding comments

1. In von Stengel and Koller (1997), it is shown that *team-maxmin equilibria are precisely the equilibria of the game with highest payoff to the team.* This observation also applies in our setting. Let $(\underline{x}, \overline{y})$ be a team-maxmin equilibrium of $G = (X_1, \dots, X_n, Y, u)$ and (x^*, y^*) any equilibrium of G. Then $u(x^*, \cdot)$ is minimized by y^*, i.e., $u(x^*, y^*) = \min_{y \in Y} u(x^*, y)$; but, since $u(\underline{x}, \overline{y}) = \max_{x \in X} \min_{y \in Y} u(x, y) \geq \min_{y \in Y} u(x^*, y)$, we have $u(\underline{x}, \overline{y}) \geq u(x^*, y^*)$ for any equilibrium (x^*, y^*) of G.

2. The equilibrium strategy of the adversary is not a minmax strategy; the minmax cost is greater than the team-maxmin payoff as long as we consider all strategy profiles of the team. However, if we restrict the set of strategy profiles X to $\hat{X} := \bigcup_{i=1}^{n} \left(X_i \times \{\underline{x}_{-i}\}\right)$, where $\underline{x}$ is a team-maxmin strategy profile, then we have a maxmin equality

$$\max_{x \in \hat{X}} \min_{y \in Y} u(x, y) = \min_{y \in Y} \max_{x \in \hat{X}} u(x, y).$$

To see this, observe that the equilibrium $(\underline{x}, \overline{y})$ of $\Gamma_1 = (X, Y, \sum_i v_i)$ is also an equilibrium of a two-player zero-sum game $\Gamma_2 = (\hat{X}, Y, u|_{\hat{X} \times Y})$, where $u|_{\hat{X} \times Y}$ is a restriction of u to $\hat{X} \times Y$. Thus, $(\underline{x}, \overline{y})$ is a pair of a maxmin and a minmax strategies of Γ_2 satisfying

$$\max_{x \in \hat{X}} \min_{y \in Y} u(x, y) = u(\underline{x}, \overline{y}) = \min_{y \in Y} \max_{x \in \hat{X}} u(x, y).$$

3. Research on discontinuous games—and more broadly, generalized games with discontinuous payoffs—is an active area of game theory with application to general equilibrium theory (He and Yannelis 2017; Anderson et al. 2022; Podczeck and Yannelis 2024; Khan et al. 2024, 2025). Whether the results of this paper extend to such games is an open question. In particular, we want to pose the following question: Can we replace the concave–convex assumption of Theorem 3.1 with a *quaiconcave–quasiconvex* one? This would allow highly discontinuous payoffs. One way of looking at this problem would be to consider the collection of two-player zero-sum games between a member i of the team and the adversary, $\Gamma^i = (X_i, Y, v_i)$, $i = 1, \dots, n$, where v_i is a function defined in the proof of Theorem 3.1. Even under the quasiconcave–quasiconvex assumption, by Sion's theorem, the maxmin equality $\max_{x_i} \min_y v_i(x_i, y) = \min_y \max_{x_i} v_i(x_i, y)$ holds for each game, with maxmin

strategies $\underline{x}_i$ coinciding with the srategies of team-maxmin strategy profile, and minmax strategies $\overline{y}^i$ of the adversary, one for each Γ^i. Thus, from this viewpoint, the question will be: Can the adversary choose a *common* minmax strategy $\overline{y}$? Under the concave–convex assumption, the answer is yes; under the quasiconcave–quasiconvex assumption, it remains open.

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Data availability Data sharing is not applicable. We do not analyse or generate any datasets, because our work proceeds within a theoretical and mathematical approach.

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